

INTERNAL ADVISORY MEMORANDUM

Protective Put Strategy — Downside Hedge for Large-Cap Equity Holding

To: Portfolio Manager / Mentor
From: Junior Derivatives Analyst, Wealth Desk
Date: 28 May 2026
Subject: Options Hedge — 500 shares @ ₹2,450 | 3-Month Horizon
Classification: Internal | Week 1 Deliverable

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1. Client Situation

Position Summary

Our client holds 500 equity shares of a large-cap Indian stock currently trading at ₹2,450 per share on the NSE, giving a total market exposure of ₹12,25,000. The client is concerned about a potential correction of 10-15% (implying a spot fall to ₹2,082-₹2,205) over the next three months (approx. 91 days).

Binding Constraint

The client is unwilling to liquidate the holding due to two considerations:

- Tax efficiency: Sale would trigger capital gains tax under the Income Tax Act, 1961. LTCG on listed equity above ₹1 lakh is taxed at 12.5% (post Budget 2024); crystallising gains unnecessarily is sub-optimal.
- Fundamental conviction: The client believes the stock will recover and wishes to retain long-term upside.

Objective

Design a downside hedge using exchange-traded options on NSE that caps losses below a chosen floor price while preserving full upside participation, within a defined premium budget.

Parameter	Value
Current Spot (S0)	₹2,450
Shares Held	500
Total Exposure	₹12,25,000
Hedge Horizon	3 months (approx. 91 days)
Downside Concern	10-15% correction (₹2,082-₹2,205)
Constraint	No sale of underlying shares

2. Call vs Put — Which Option Type Fits?

Payoff Definitions

An option is a contract granting the buyer the right — but not the obligation — to transact the underlying at a pre-agreed strike price K on (or before) expiry.

Option Type	Payoff Formula	Buyer's View	Fits Our Need?
Call (Long)	$\max(ST - K, 0)$	Bullish — profits if ST rises above K	No
Put (Long)	$\max(K - ST, 0)$	Bearish/Hedge — profits if ST falls below K	Yes

Why a Long Put?

The client's risk is a falling stock price. A long put pays off when $ST < K$, directly offsetting equity losses. A call pays off when $ST > K$ — the opposite direction — and therefore provides no downside protection.

Numeric Examples ($K = ₹2,450$)

Scenario	ST	Call Payoff $\max(ST - K, 0)$	Put Payoff $\max(K - ST, 0)$
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Stock falls 12%	₹2,156	$\max(2156 - 2450, 0) = ₹0$	$\max(2450 - 2156, 0) = ₹294$
Stock unchanged	₹2,450	$\max(2450 - 2450, 0) = ₹0$	$\max(2450 - 2450, 0) = ₹0$
Stock rises 10%	₹2,695	$\max(2695 - 2450, 0) = ₹245$	$\max(2450 - 2695, 0) = ₹0$

The put pays ₹294 per share when the stock drops 12%, compensating the equity loss. The call provides zero protection — confirming that a long put is the correct instrument.

3. Strike Selection & Moneyness

Proposed Strike: K = ₹2,200 (OTM Put)

We recommend buying a 2,200-strike put (approx. 10% OTM). The rationale:

- Cost efficiency: An OTM put costs less premium than an ATM (₹2,450) put because it has zero intrinsic value; the entire premium is time value. This keeps the hedge budget manageable.
- Alignment with stated risk: The client fears a 10–15% fall; a floor at ₹2,200 activates precisely within that band, letting minor volatility pass unhedged.
- NSE availability: NSE lists strikes in ₹50 increments for large-cap stocks; ₹2,200 is a standard listed strike.

Moneyness Classification

Strike K	vs Spot S0 = ₹2,450	Moneyness	Intrinsic Value	Time Value
₹2,700	$K > S0$	In-the-Money (ITM)	₹250 per share	Remainder of premium
₹2,450	$K = S0$	At-the-Money (ATM)	₹0	Full premium
₹2,200	$K < S0$	Out-of-the-Money (OTM)	₹0 (our choice)	Full premium

For our ₹2,200 put: Intrinsic value = $\max(2200 - 2450, 0) = ₹0$. All premium paid (estimated ₹35–55 per share for a 3-month option, based on NSE implied vols of approx. 18–22%) is pure time value, which decays as expiry approaches (theta risk). An ATM put would offer a lower breakeven but at 2–3× the premium cost.

4. European vs American Exercise Style

Feature	European Put	American Put
Exercise timing	Only at expiry	Any time up to expiry
Premium (all else equal)	Lower	Higher (early-exercise premium)
NSE equity options	Standard for index options (Nifty)	Standard for individual stock options
Pricing model	Black-Scholes (closed form)	Binomial / numerical methods

Does Early Exercise Matter for Our Hedge?

On NSE, individual stock options are American-style (exercisable any day before expiry), while index options (Nifty, Bank Nifty) are European-style. For our stock put, early exercise could theoretically be optimal when the put is deep in-the-money and carries little remaining time value.

The Interest-Rate Argument for Early Exercise

A deep ITM American put may be exercised early when:

- The stock has fallen sharply — e.g. spot drops to ₹1,900 against $K = ₹2,200$.
- Intrinsic value (₹300) dominates; time value is negligible.
- The interest earned on the exercise proceeds ($₹2,200 \times \text{risk-free rate}$) exceeds the remaining optionality. With the RBI repo rate at approx. 6.25% (May 2026), the daily carry on ₹2,200 is approx. ₹0.38 — small but non-trivial over weeks if the option has no further downside to capture.

Practical implication: For a hedge (not speculation), early exercise is generally not recommended. Exercising early removes the insurance — if the stock bounces from ₹1,900 back to ₹2,300, you have surrendered the hedge. It is almost always better to sell the put in the market (capturing remaining time value) rather than exercise early.

Index option note: If we use Nifty/Bank Nifty puts as a proxy hedge (see Section 6), European-style exercise eliminates early-exercise risk entirely — a useful simplifying advantage.

5. Payoff Table at Expiry

Assumptions: $K = ₹2,200$ | Premium paid (p) = ₹45/share | 500 shares held | No transaction costs.
Portfolio outcome per share = Stock P&L; + Put payoff – Premium paid. Multiply by 500 for total portfolio figures.

ST at Expiry	Stock P&L (ST – 2450)	Put Payoff $\max(2200 - ST, 0)$	Net Option (Payoff – 45)	Portfolio per share	Total ₹ (× 500)
₹2,100	–₹350	₹100	+₹55	–₹295	–₹1,47,500
₹2,200	–₹250	₹0	–₹45	–₹295	–₹1,47,500
₹2,300	–₹150	₹0	–₹45	–₹195	–₹97,500
₹2,450	₹0	₹0	–₹45	–₹45	–₹22,500
₹2,600	+₹150	₹0	–₹45	+₹105	+₹52,500
₹2,700	+₹250	₹0	–₹45	+₹205	+₹1,02,500

Key Observations

- Maximum loss is capped at –₹295/share (–₹1,47,500 total) for any $ST \leq ₹2,200$ — the put floor is working. Without the hedge, a fall to ₹2,100 would cost ₹1,75,000.
- Break-even (with hedge) = ₹2,450 – ₹45 = ₹2,405. The stock must stay above ₹2,405 at expiry for the hedged portfolio to show a net profit.
- Upside is fully preserved (less the ₹45 premium sunk cost): at ₹2,700 the client nets +₹1,02,500 vs +₹1,25,000 unhedged — only ₹22,500 forgone.

6. Recommendation

Proposed Trade

Parameter	Detail
Strategy	Long Protective Put
Underlying	Individual stock (NSE-listed large-cap)
Action	BUY 5 put contracts (lot size 100 shares each = 500 shares total)
Strike (K)	₹2,200 (approx. 10.2% OTM)
Expiry	August 2026 monthly expiry (approx. 91 days)
Est. Premium	₹40–55 per share Total outlay approx. ₹20,000–₹27,500
Max Loss (hedged)	₹1,47,500 (vs ₹1,75,000 unhedged at $ST = ₹2,100$)
Insurance benefit	Up to ₹27,500 in extra loss avoided at deep correction

Alternative: Index Proxy Hedge (Nifty Put)

If stock-specific options are illiquid or bid-ask spreads are wide, a Nifty 50 or Bank Nifty put can serve as a proxy hedge:

- Uses European-style options — no early-exercise complexity.
- Benefits from deeper liquidity and tighter spreads on NSE F&O.;
- Introduces basis risk: the stock may not move 1-for-1 with the index; effectiveness depends on the stock's beta (ideally above 0.9).

Reference: Zerodha Varsity, Hedging with Options — <https://zerodha.com/varsity/chapter/hedging-with-options/> — covers the protective put structure and basis risk in an Indian market context.

Key Risks

- Theta (time) decay: The ₹45 premium erodes daily. Decay accelerates in the final 30 days before expiry.
- Wrong strike: If the stock corrects only 8% (to ₹2,254), the ₹2,200 put expires worthless and the entire ₹22,500 outlay is lost.
- Liquidity risk: OTM individual-stock strikes can carry wide bid-ask spreads. Prefer strikes with open interest above 10,000 contracts.
- Implied volatility crush: A fall in IV after purchase reduces the put's market value even if spot is unchanged (vega risk).
- Lot-size mismatch: NSE lot sizes are fixed and may not perfectly match 500 shares, leaving a residual unhedged gap.

7. Reflection

Before working through the payoff explorer and the Varsity modules on options, I intuitively thought of hedging as 'selling something to offset risk' — which would mean selling the shares or shorting futures. The course reframed this entirely: a long put is insurance, not a negative bet. You pay a defined, upfront premium and retain every rupee of upside if your fears prove unfounded. The payoff diagram — a flat floor to the left of the strike and an unimpeded slope to the right — made the asymmetry viscerally clear in a way the formula alone did not. I also underestimated how much time value dominates OTM option pricing: an option with zero intrinsic value still commands a meaningful premium purely because of the probability distribution of future prices. This changed how I think about premium budgets — the cost is not just the strike distance, it is the market's pricing of uncertainty over time. The ATM vs OTM trade-off taught me that hedging is a continuous decision, not a binary one: tighter protection costs more, and the 'right' strike depends on how much premium drag the portfolio can absorb versus how much downside risk the client is truly unwilling to bear.

References

- [1] Zerodha Varsity — Hedging with Options. <https://zerodha.com/varsity/chapter/hedging-with-options/> (accessed May 2026).
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- [4] SEBI Circular SEBI/HO/MRD/DP/CIR/P/2019/65 — Review of Margin Framework for Options. Securities and Exchange Board of India, 2019.
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